

Camilo Becerra

69 Brown St, Box #4376 Providence, RI 02912 | Phone: (401) 636-4082 | camilo_becerra@brown.edu | [LinkedIn](#) | [Projects](#)

EDUCATION

Brown University, B.Sc. Applied Mathematics-Computer Science Providence, United States | **Expected Graduation May 2027**

- 4.0/4.0 GPA
- Relevant Coursework: *Computer Vision, Operating Systems, Accelerated Introduction to Computer Science (Data Structures and Algorithms), Operations Research: Probabilistic Models, Linear Algebra, Multivariable Calculus, Applied Ordinary Differential Equations, Statistical Inference I.*
- Activities and Societies: Statistical Inference I Teaching Assistant, Basic Spanish II Teaching Assistant, MAMBO Latin American Orchestra Co-Founder and Co-President, Colombian Student Association Financial Signatory.

PROJECTS

Football Real-Time Game Analysis, Python, YOLO, OpenCV December 2024

- Fine-tuned YOLO v11 model using the Roboflow football player detection dataset, achieving 0.35 classification loss.
- Designed and implemented a team clustering and detection algorithm based on the K-means clustering algorithm.
- Used results from the YOLO object detections to achieve object tracking per frame for the ball and players from both teams.

Computer Vision Toolkit, Python, Numpy, Scikit-Image September 2024 – November 2024

- Developed computer vision tools for image filtering, feature matching, 3D reconstruction, and scene classification tasks.
- Implemented various classical computer vision algorithms: the convolution operation iteratively and through the Fast Fourier Transform, the Harris corner detector, a SIFT-like feature descriptor, RANSAC, bag of words, and one vs. all linear SVM
- Fine-tuned VGG-F deep learning model for scene classification, achieving 85% accuracy on validation data.

Financial Tools Web Application, Python, Matplotlib, Streamlit, Alpha Vantage API August 2024

- Developed an option pricing module employing the Black-Scholes formula to calculate premiums and Greeks, for more informed decision-making when options trading.
- Created an implied volatility calculator by integrating the Black-Scholes model and the Newton-Raphson algorithm to derive volatility in market prices.
- Built a comprehensive P&L analysis tool that visualizes various P&L scenarios using a heatmap and a line graph, based on the movement of underlying asset prices and volatility.
- Designed a real-time stock analysis tool displaying stock quotes and 30-day candlestick using the Alpha Vantage API.

RESEARCH EXPERIENCE

Brown University, Department of Physics Research Assistant Providence, United States | December 2024 – Present

- Developed knowledge distillation on multilayer perceptrons for top jet tagging using the ATLAS dataset, achieving AUC of 0.94 for teacher and 0.92 for student models on ROC curves while achieving a 90x size reduction.
- Collaborate with professors and postdoctoral fellows at Brown and CERN to develop fast-inference techniques for classification tasks to analyze jets created by proton collisions in the Large Hadron Collider in Geneva, Switzerland.

WORK EXPERIENCE

National Student Leadership Conference, Team & Assistant Team Advisor Johns Hopkins University, USA | June 2024 – July 2024

- Led lectures, labs, and workshops in deep learning, regression and classification algorithms, data analysis, data presentation, data cleaning, and other topics in AI and data science to groups of up to 30+ students at a time.
- Mentored students in the development of projects in AI and data science, namely developing machine learning models or analyzing large datasets to extract insights.
- Staffed 4 summer sessions, receiving a total of over 800 students, with around 150 students in my specific programs.

LEADERSHIP EXPERIENCE

MAMBO Latin American Orchestra @ Brown University, Co-Founder Providence, RI | September 2024 – Present

- Founded the only space for undergraduate and graduate Brown and RISD students to rehearse and perform various Latin American genres including salsa and merengue.
- Organize weekly rehearsals for a team of 10 members to practice songs of our choice to perform for the Brown community.
- Search for and schedule performances for the orchestra to share our music with the community and improve musically.
- Design advertisements to promote the orchestra's activities, including performances, recruiting, and open rehearsals.

Colombian Student Association @ Brown University, Executive Board Member Providence, RI | January 2024 – Present

- Organize cultural, social, and academic activities and trips centered around Colombian culture, open to all students.
- Serve as financial signatory for the organization, ensuring compliance with the rules that govern the association's finances.

SKILLS

Technical Skills: Python, C/C++, Java, HTML/CSS, JavaScript, React, MySQL, Firebase, AWS, Linux, Git, Microsoft Office

Languages: English (Native), Spanish (Native), Japanese (Intermediate)

Hobbies/Interests: Playing the saxophone and guitar, cooking and baking, sports (tennis, basketball, soccer, golf).